

Adioz D Eshitemi

Full-stack AI Engineer | Agentic AI in Production | Rust · Java · Python · Go · Angular
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PROFESSIONAL SUMMARY

Full-stack AI engineer with 8 years of experience in backend systems and web applications. Takes AI into production for enterprise clients and owns each system from design to support. Builds tool-using agents that make real and auditable changes. Trains and distills models on GPU, then serves them on CPU. Believes AI should be as efficient as it is intelligent, so designs resource-aware AI: small, quantized models that meet tight latency budgets without wasted compute. Treats agents as production software, with typed tools, fail-closed guardrails and a named operator behind every change.

TECHNICAL SKILLS

- **Agentic AI:** Tool-using and ReAct agents/evaluation, typed audited tools, retrieval (RAG) & local SLMs.
- **ML Engineering:** Burn, Candle, PyTorch, CLIP, embeddings, model training and evaluation.
- **Resource-Aware AI:** Multi-teacher distillation into small students, INT8 quantization, truncatable embeddings (MRL), CPU-only inference, zero-copy memory-mapped weights, SIMD scoring, cached teacher passes, accuracy-driven model sizing, latency and memory budgets, load profiling.
- **Backend:** Rust (Tokio, Axum), Java (Spring Boot), Golang (gRPC), Python, REST, microservices.
- **Frontend:** Angular, TypeScript, RxJS, Next.js, Rust/WebAssembly, HTML5/SCSS.
- **Data:** PostgreSQL, MongoDB, SQLite, ClickHouse, Kafka, Redis, Tantivy (full-text search), Qdrant.
- **DevOps & Security:** Kubernetes, AWS, GCP, GitHub Actions, CI quality gates, JWT, OAuth2, RBAC and Offensive Rust (Exploit PoCs, Custom Security Tooling, Low-Level Memory Security).

PROFESSIONAL EXPERIENCE

Lead Engineer (AI & Backend) | [BBR \(Big Bold Red Agency\)](#) | 2025 – Present

- **End-to-end ownership:** Own BBRaiN across its Rust inference service, vendor hub, admin console and GPU training pipeline; deployed inside each client's own cloud (VPC).
- **Agentic AI in production:** Built an admin assistant agent that turns plain-language requests into audited configuration changes through typed write tools.
- **AI in the binary:** Serve vision, embedding and ranking models from a single Rust service on CPU, with no GPU runtime, within a tight latency budget.
- **Guardrails and licensing:** Built fail-closed content safety controls, per-feature licensing, and custom security validation tooling in Rust (Offensive Rust methodologies) to prevent adversarial manipulation and unauthorized feature execution.

Junior AI Engineer | [Berrijam](#) | 2023 – 2025

- **AI integration:** Integrated ML models and LLM APIs into Go, Java and Python microservices over REST and gRPC.
- **LLM features:** Built prompt, retrieval (RAG) and structured-output features for client-facing tools.
- **Data preparation:** Cleaned, transformed and validated client datasets for modelling and analytics.
- **Prototype to production:** Turned validated AI prototypes into services with Docker, CI pipelines and monitoring.
- **Evaluation and docs:** Tested model outputs against baselines and documented APIs.

Fullstack Engineer (Java & Go) | [Zone01](#) | 2023 – 2025

- **Service performance:** Cut inter-service overhead by 30% by moving services to Golang and gRPC.
- **Streaming pipelines:** Ran Kafka pipelines with sub-second ingestion for real-time monitoring.
- **Team standards:** Set code-quality and DevSecOps standards and mentored engineers.

Backend Engineer (Golang & Java) | [Upwork](#) | 2018 – 2024

- **Real-time systems:** Built Golang (REST, WebSockets) and Java Spring Boot backends for international startups.
 - **Full-stack delivery:** Integrated Angular frontends with Go and Java backends through well-documented APIs.
 - **Reliability:** Deployed Dockerized services with auto-scaling and monitoring, holding 99% uptime.
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CURRENT CORE RESPONSIBILITIES — BBRaiN AI

Agentic AI: Admin Assistant | Rust · text encoder · slot extraction · typed tools

- **Requests to actions:** Extend the agent that turns plain-language requests into changes to pages, rules, configurations and feeds for complex data flows.
 - **Safe by design:** Makes auditable changes behind guardrails that roll back when the agent is unsure or a change fails its gates.
 - **Evaluation:** Test the assistant on scripted operator conversations and fix what it misunderstands.
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Model Training & Distillation | Burn · PyTorch · GPU · CLIP ViT-B/32

- **Distillation:** Distil large video and audio teacher models into a compact student model.
 - **Model types:** Train a content encoder, content-profile heads, a behavioural projector, a small reasoning core with learned priors, the assistant's text encoder and a vision model.
 - **Evaluation:** Test each model on held-out data and track how data balance sets the accuracy ceiling.
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Data & Model Release Pipelines | safetensors · MovieLens · INT8 · Ed25519

- **Training data:** Cache teacher outputs and corpus once on GPU so training never reloads the teachers, and ingest public interaction datasets into normalized event records.
 - **Signed releases:** Quantize models and sign them so the service loads only verified bundles.
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Serving, Search & Hot Swapping | Rust · Axum · Tantivy · Rhai

- **Hybrid search:** Blend BM25 keyword and semantic search with personalized ranking efficiently.
 - **Hot swapping:** Run business rules as Rhai scripts that swap at runtime, without a deployment.
 - **Profiling:** Profile release builds under load, find the cause of latency regressions and fix them.
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Delivery & Operations | CI gates · Postman · Docker · client VPCs

- **Design to release:** Design, build and test each change, then pass the CI gate (lint, tests, doc checks).
 - **API contracts:** Keep Postman, samples and the client apps in step with every route change.
 - **Load and resilience testing:** Run smoke, back-pressure, spike and resource-consumption tests.
 - **Audits and support:** Run security and reliability audits and track each finding to closure.
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KEY PROJECTS

BBRaiN — AI search, recommendations and personalization | Rust · Axum · Burn · CLIP · Tantivy

- **Product:** [One AI engine](#) for search, recommendations and personalization across all media.
 - **In production:** Powers content discovery on [Baze streaming platform](#).
 - **Playground:** [Viewer app](#) used for pitching and [CMS studio](#); access key available on request.
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Hisiya — AI sentiment intelligence for African markets

- **Local languages:** Reads brand sentiment in Swahili, Sheng, Pidgin and Amharic languages.
 - **Brand intelligence:** Turns conversations into a brand-health score, trends and benchmarks.
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CERTIFICATIONS & CONTINUOUS LEARNING

- [ALX Africa](#) – Software Engineering (2021 – 2024)
 - [Zone01 Kisumu](#) – Apprentice Mentorship (2024 – 2025): Intensive software engineering hands-on mentorship in an enterprise setup. Credential Z01P-301.
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HACKATHONS & COMMUNITY

- [Agentic AI Hackathon](#) (Jan 2026, 48 h): Built a ReAct agent for remote farmers that takes spoken requests in Swahili, diagnoses plant disease from a photo and alerts agricultural officers.
- **Sui Foundation Community** – Developer Event (Dec 2025): Hands-on Sui blockchain workshop.
- **Kisumu Tech Community** – Member (06/2024 – Present)
- **Nairobi Gophers** – Member (04/2024 – Present)